DATA ITEM DESCRIPTION

Title: MIL-STD-3023A HARDNESS ALLOCATION REPORT (HAR)

Number: DI-EMCS-81850A
AMSC Number: 10411
DTIC Applicable: No
Preparing Activity: DS

Approval Date: 20230320
Limitation: N/A
GIDEP Applicable: No
Project Number: EMCS-2023-007

Use/Relationship:

The MIL-STD-3023A Hardness Allocation Report (HAR) applies to MIL-STD-3023A*, High-Altitude Electromagnetic Pulse (HEMP) Protection for Military Aircraft*. DI-EMCS-81850, *Technical Report for MIL-STD-3023 Hardness Allocation*, is hereby cancelled and superseded by DI-EMCS-81850A. This data item description (DID) contains the format, content, and intended-use information for the data deliverable resulting from the hardness allocation procedure reporting tasks performed in accordance with Appendix A of MIL- STD-3023A.

The HAR is required to document the hardness allocation procedure. This report serves as an evolving document that verifies the hardness allocation as the aircraft engineering development and production programs develop. The initial HAR starts as a paper study, but is then updated and enhanced through the aircraft development cycle based on test data generated from test programs outlined in Appendices B, C, and D of MIL-STD-3023A. This report initially needs to be submitted for government approval at an aircraft program's preliminary design review (PDR). Subsequent submittals should be scheduled around updates in the allocation procedure as identified by the various test programs used to verify the actual hardness allocation. The HAR should include engineering change records, hardware modification records, engineering judgment records (EJRs), service experience records, and other related data.

This DID supersedes DI-EMCS-81850.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and any applicable amendments, notices, and revisions, shall be as specified in the contract. The HAR's classification shall be in accordance with any relevant aircraft-specific or system-specific classification guides and also the Defense Threat Reduction Agency (DTRA) Security Classification Guide for DoD Electromagnetic Pulse (EMP) Programs and Activities. (DTRA's security classification guide is available by mail request to ATTN: RD-NTSA, Defense Threat Reduction Agency, 8725 John J. Kingman Road, MSC 6201, Fort Belvoir, VA 22060-6201.)

2. Format. Contractor format is acceptable for report (comma-separated values (CSV) is the preferred data format).

DISTRIBUTION STATEMENT A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:54Z
Check the source to verify that this is the current version before use.

3. Content. The HAR documents the hardness allocation procedure for protecting military aircraft from HEMP. The HAR shall include empirical and analytical data generated by the allocation procedure described in detail in Appendix A of MIL-STD-3023A.

3.1 Administrative data. The HAR shall contain an administrative section covering the following:

a. Contract number, contract data requirements list (CDRL) number, and date

b. Authentication and certification by a qualified representative of the procuring activity for the HAR

3.2 Summary information. The HAR shall be presented in the style of the technical report which is recommended in Appendix A of MIL-STD-3023A. It shall include the aircraft hardness concept, electromagnetic (EM) shielding barriers, point of entry (POE) definitions, and performance requirements. This report shall be updated as experimental data from Appendices B, C, and D of MIL-STD-3023A become available to ensure that the design margin specifications; EM Barrier, MCS Immunity; and reasonable worse case current requirements for the aircraft are met with reasonable confidence. In addition to the written technical report, the data obtained from testing associated to the hardness allocation procedure shall be presented in the contractor's electronic format and contain all data described in Appendix A of MIL-STD-3023A. The electronic data format will be described in the written report. Appropriate use of appendices for large amounts of information with a short summary of key facts in the main text is encouraged.

The HAR shall summarize the following:

3.2.1 Introduction, including:

a. General objectives of the HAR

b. A general description of the hardness allocation procedure

c. A brief overview of the aircraft hardness design concept

d. A statement of any assumptions or limitations associated with the hardness allocation procedure or associated test data

3.2.2 Scope. The HAR shall provide results and technical data developed for each of the seven steps of the hardness allocation procedure as defined in Appendix A, paragraphs A.4 and A.5, of MIL-STD-3023A. The report outline, as detailed in Appendix A, paragraph A.5.7, of MIL-STD-3023A includes the following topics:

a. A complete schedule of EM barriers, their performance requirements, and the lists of equipment both within each barrier and outside all barriers

b. A complete schedule of penetrations for each EM barrier with their performance requirement, to include, for each POE, the associated hardness allocation estimate, the test requirements, and the reasonable worst-case current for each POE

c. POE penetration protection device (PPD) tests and test results

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:54Z
Check the source to verify that this is the current version before use.

d. Production-level mission critical systems (MCS) immunity test (PMIT) results

e. HEMP protection subsystem (HPS) EM barrier test results, to include descriptions of the test point selections

f. HPS immunity test results

g. HEMP verification test results, to include the active system test (AST), internal direct drive test (DDT), and external DDT results

3.2.3 Schedule of HAR submittal. The HAR is a living instrument that documents the hardness allocation as the aircraft engineering design program and production proceeds. There are at least five major submittals as fully defined in Appendix A, paragraph A.5.7.2, of MIL-STD-3023A, listed as follows:

a. Initial submittal by the preliminary design review (PDR) for government approval, to contain descriptions of the hardness concept; the barrier definitions with the lists of equipment within the barriers and equipment exterior to barrier; and the schedule of penetrations for each barrier with the associated test requirements.

b. Second submittal after the critical design review (CDR). All hardness allocation details should be known and presented.

c. Third submittal after the production hardness assurance (PHA) testing (but not the quality assurance portion) is complete, to contain descriptions of the type of tests performed; the test results relative to the test requirement in the initial submittal; and, if deficiencies exist, plans for addressing each deficiency and plans for subsequent tests to demonstrate HEMP hardness compliance.

d. Fourth submittal after the HPS performance testing is complete, to contain all POE evaluation test results and the results for the MCS inside the barrier where the transient currents on the MCS interface connectors were measured.

e. Fifth submittal after the first LRIP system verification testing has been completed, to contain the final verification that the internal currents at each MCS interface are less than the MCS immunity by the design margin, and a statement that no mission critical failures were observed based on the active system testing.

The HAR shall be revised during fleet production upon any verification test failures and any modification of the HEMP protection features of the aircraft. The exact details of the failure on the specific aircraft and future production aircraft is at the discretion of the operating or procuring agency, but shall be documented as part of the revised HAR.

End of DI-EMCS-81850A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:54Z
Check the source to verify that this is the current version before use.